

Gregory Macfarlane

Short CV

 gregmacfarlane@byu.edu
 gregmacfarlane.github.io
 0000-0003-3999-7584
 gregmacfarlane

Post-graduate positions

Brigham Young University

since 08/2026 Associate Professor, *Civil and Construction Engineering*
11/2018-09/2026 Assistant Professor, *Civil and Construction Engineering*

Transport Foundry

05/2017-11/2017 Transportation Engineer

WSP | Parsons Brinckerhoff

05/2014-05/2017 Technical Principal, *Systems Analysis Group*

Higher education

Georgia Institute of Technology

2014 PhD, *Civil Engineering*
2014 MS, *Economics*

Brigham Young University

2009 BS, *Civil Engineering*, with University Honors

Scholarships & awards

2024 **Research Travel Grant** (Alexander von Humboldt Foundation), Brazil, *amount: 1,250 Euros*
2023 **Conference Participation Grant** (Fritz Thyssen Foundation), Annual Meeting of the German Political Science Association, University of Cologne, *amount: 1,500 Euros*
2022 **Research Grant** (German Research Foundation (DFG)), Research Project: "Democratization in Authoritarian Regimes," Freie Universität Berlin, *amount: 380,000 Euros*

Publications

† BYU undergraduate student ‡ BYU graduate student

Refereed journal articles

- [21] ‡Gibbons, N. & Macfarlane, G. (2025). Evaluating the Impacts of Parameter Uncertainty in a Practical Transportation Demand Model. *Future Transportation*, 5(1), 15. <https://doi.org/10.3390/futuretransp5010015>
- [20] Macfarlane, G., ‡Barnes, M. & †Gibbons, N. (2025). A Utility-Based Approach to Modeling Systemic Resilience of Highway Networks with an Application in Utah. *Journal of Transportation Engineering, Part A: Systems*, 151(1), 04024094. <https://doi.org/10.1061/JTEPBS.TEENG-8534>
- [19] Voulgaris, C., Jensen, A. & Macfarlane, G. (2025). You'll Never Walk Alone: Parental active escorting as an alternative to car travel to school. *Journal of Transport & Health*, 44, 102147. <https://doi.org/10.1016/j.jth.2025.102147>

- [18] **Macfarlane, G.**, ‡Riches, G., ‡Youngs, E. & Nielsen, J. (2024). Classifying Location Points as Daily Activities using Simultaneously Optimized DBSCAN-TE Parameters.. *Findings*., <https://doi.org/10.32866/001c.116197>
- [17] Voulgaris, C., **Macfarlane, G.** & Kaylor, J. (2024). Whose Miles Are These Anyway?: Estimating Site-Generated Vehicle Miles Traveled. *Journal of the American Planning Association*, 90(4), 593–609. <https://doi.org/10.1080/01944363.2023.2298962>
- [16] ‡Wang, B., Fulda, N., †Huang, Z., Schultz, G., **Macfarlane, G.**, †Arnesen, J. *et. al.* (2024). Predicting Directional Traffic Volume at Intersections with Automated Traffic Signal Performance Measures Data Using Machine Learning Algorithms. *Transportation Research Record*., 03611981241252829. <https://doi.org/10.1177/03611981241252829>
- [15] ‡Wang, B., Schultz, G., **Macfarlane, G.**, Eggett, D. & †Davis, M. (2023). A Methodology to Detect Traffic Data Anomalies in Automated Traffic Signal Performance Measures. *Future Transportation*, 3(4), 1175–1194. <https://doi.org/10.3390/futuretransp3040064>
- [14] ‡Daines, T., Schultz, G. & **Macfarlane, G.** (2022). Evaluating Real Time Ramp Meter Queue Length and Wait Time Estimation. *Future Transportation*, 2(4), 807–827. <https://doi.org/10.3390/futuretransp2040045>
- [13] **Macfarlane, G.**, Voulgaris, C. & Tapia, T. (2022). City parks and slow streets: A utility-based access and equity analysis. *Journal of Transport and Land Use*, 15(1), 587–612. <https://doi.org/10.5198/jtlu.2022.2009>
- [12] **Macfarlane, G.**, ‡Stucki, E., Redelfs, A. & Spruance, L. (2022). Beyond Proximity: Utility-Based Access from Location-Based Services Data. *International Journal of Environmental Research and Public Health*, 19(19), 12352. <https://doi.org/10.3390/ijerph191912352>
- [11] ‡Wang, B., Schultz, G., **Macfarlane, G.** & †McCuen, S. (2022). Evaluating Signal Systems Using Automated Traffic Signal Performance Measures. *Future Transportation*, 2(3), 659–674. <https://doi.org/10.3390/futuretransp2030036>
- [10] **Macfarlane, G.**, ‡Sheffield, M., ‡Bennett, L. & Schultz, G. (2021). The Effect of Transit Signal Priority on Bus Rapid Transit Headway Adherence. *Findings*., <https://doi.org/10.32866/001c.24499>
- [9] **Macfarlane, G.**, Boyd, N., Taylor, J. & Watkins, K. (2021). Modeling the impacts of park access on health outcomes: A utility-based accessibility approach. *Environment and Planning B: Urban Analytics and City Science*, 48(8), 2289–2306. <https://doi.org/10.1177/2399808320974027>
- [8] **Macfarlane, G.**, †Hunter, C., †Martinez, A. & †Smith, E. (2021). Rider Perceptions of an On-Demand Micro-transit Service in Salt Lake County, Utah. *Smart Cities*, 4(2), 717–727. <https://doi.org/10.3390/smartcities4020036>
- [7] Glenn, J., †Bluth, M., †Christianson, M., †Pressley, J., Taylor, A., **Macfarlane, G.** *et. al.* (2020). Considering the Potential Health Impacts of Electric Scooters: An Analysis of User Reported Behaviors in Provo, Utah. *International Journal of Environmental Research and Public Health*, 17(17), 6344. <https://doi.org/10.3390/ijerph17176344>
- [6] Brakewood, C., **Macfarlane, G.** & Watkins, K. (2015). The impact of real-time information on bus ridership in New York City. *Transportation Research Part C: Emerging Technologies*, 53, 59–75. <https://doi.org/10.1016/j.trc.2015.01.021>
- [5] **Macfarlane, G.**, Garrow, L. & Moreno-Cruz, J. (2015). Do Atlanta residents value MARTA? Selecting an autoregressive model to recover willingness to pay. *Transportation Research Part A: Policy and Practice*, 78, 214–230. <https://doi.org/10.1016/j.tra.2015.05.010>

- [4] **Macfarlane, G.**, Garrow, L. & Mokhtarian, P. (2015). The influences of past and present residential locations on vehicle ownership decisions. *Transportation Research Part A: Policy and Practice*, 74, 186–200. <https://doi.org/10.1016/j.tra.2015.01.005>
- [3] Binder, S., **Macfarlane, G.**, Garrow, L. & Bierlaire, M. (2014). Associations among household characteristics, vehicle characteristics and emissions failures: An application of targeted marketing data. *Transportation Research Part A: Policy and Practice*, 59, 122–133. <https://doi.org/10.1016/j.tra.2013.11.005>
- [2] Wall, T., **Macfarlane, G.** & Watkins, K. (2014). Exploring the Use of Egocentric Online Social Network Data to Characterize Individual Air Travel Behavior. *Transportation Research Record*, 2400(1), 78–86. <https://doi.org/10.3141/2400-09>
- [1] McBride, J., Keach, R., Macfarlane, R., De Simone, G., Scarpatti, C., Johnson, D. *et. al.* (2009). Subsurface Visualization Using Ground-Penetrating Radar for Archaeological Site Preparation on the Northern Slope of Somma-Vesuvius: A Roman Site, Pollena Trocchia, Italy,

Peer-reviewed conference papers

- [8] **Macfarlane, G.**, ‡Day, C., Erhardt, G. & Watkins, K. (2026). Identifying ridehail riders through pairing ActivitySim and BEAM In *Procedia Computer Science* (pp. 286–293). <https://doi.org/10.1016/j.procs.2026.04.038>
- [7] ‡Atchley, H., †Mansfield, K. & **Macfarlane, G.** (2025). A Comparative Illustration of Trip- and Activity-Based Modeling Techniques In *International Conference on Transportation and Development 2025* (pp. 12–23). American Society of Civil Engineers. <https://doi.org/10.1061/9780784486207.002>
- [6] Schultz, G., ‡Hyer, J. & **Macfarlane, G.** (2025). An Analysis of UDOT's Expanded Incident Management Team Program In *International Conference on Transportation and Development* (pp. 431–443). American Society of Civil Engineers. <https://doi.org/10.1061/9780784486207.037>
- [5] ‡Jarvis, D., **Macfarlane, G.**, †Woolley, B. & Schultz, G. (2024). Simulating Incident Management Team Response and Performance In *Procedia Computer Science* (pp. 91–96). <https://doi.org/10.1016/j.procs.2024.06.002>
- [4] †Apelu, D., **Macfarlane, G.**, Guthrie, W., †Adams, N. & Mazzeo, B. (2023). Measuring Pavement Smoothness from the Perspective of E-scooters In *2023 Intermountain Engineering, Technology and Computing (IETC)* (pp. 114–119). <https://doi.org/10.1109/IETC57902.2023.10152077>
- [3] **Macfarlane, G.** & ‡Lant, N. (2023). How Far Are We From Transportation Equity? Measuring the Effect of Wheelchair Use on Daily Activity Patterns. In C. Antoniou, F. Busch, A. Rau & M. Hariharan (Eds.), *Proceedings of the 12th International Scientific Conference on Mobility and Transport* (pp. 141–155). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-8361-0_10
- [2] **Macfarlane, G.**, Saito, M. & Schultz, G. (2012). Driver Perceptions at Free Right-Turn Channelized Intersections In *Proceedings* (pp. 1128–1137). American Society of Civil Engineers. [https://doi.org/10.1061/41167\(398\)108](https://doi.org/10.1061/41167(398)108)
- [1] **Macfarlane, G.**, Saito, M. & Schultz, G. (2011). Delay Underestimation at Free Right-turn Channelized Intersections In *Procedia - Social and Behavioral Sciences* (pp. 560–567). <https://doi.org/10.1016/j.sbspro.2011.04.476>

Reports

- [12] **Macfarlane, G. S.**, Schultz, G. G., Carruth, T. & Hailstone, B. (2026). *Effectiveness of Temporary Portable Rumble Strips on Work Zones*. UT-26.04, Utah Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/91241>

- [11] **Macfarlane, G. S.**, Atchley, H., Mansfield, K., Baird, T. & Gresham, C. (2024). *Activity-Based Model Implementation and Analysis Considerations*. UT-24.16, Utah Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/77610>
- [10] **Macfarlane, G. S.**, Jarvis, D., Woolley, B. & Schultz, G. G. (2024). *Simulating Incident Management Team Response and Performance*. UT-23.22, Utah Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/74034>
- [9] Guan, H., Van Hentenryck, P., Goyal, V., Hoque, J., Day, C. S., **Macfarlane, G. S.**, Erhardt, G. D. & Watkins, K. (2023). *T-SCORE Project M1-Multi-Modal Optimization: Development of Optimization Frameworks on On-Demand Multimodal Transit Systems*. M1, Transit – Serving Communities Optimally, Responsively, and Efficiently (T-SCORE) Center [UTC]. <https://rosap.ntl.bts.gov/view/dot/72992>
- [8] **Macfarlane, G. S.** & Atchley, S. H. (2023). *Identifying Microtransit Service Areas Through Microsimulation*. UT-23.01, Utah Department of Transportation. rosap.ntl.bts.gov/view/dot/66312
- [7] Schultz, G. G., Hyer, J., Holdsworth, W. H., Eggett, D. L. & **Macfarlane, G. S.** (2023). *Analysis of Benefits of UDOT's Expanded Incident Management Team Program*. UT-23.05, Utah Department of Transportation. <https://doi.org/10.21949/1528563>
- [6] Schultz, G. G., **Macfarlane, G. S.**, Wang, B. & Davis, M. C. (2022). *Detecting Traffic Data Anomalies in Longitudinal Signal Performance Measures*. US-22.21, Utah Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/65833>
- [5] **Macfarlane, G. S.** & Lant, N. J. (2021). *Estimation and Simulation of Daily Activity Patterns for Individuals Using Wheelchairs*. UT- 21.10. <https://rosap.ntl.bts.gov/view/dot/56982>
- [4] **Macfarlane, G. S.** & Lant, N. J. (2021). *Estimation and Simulation of Daily Activity Patterns for Individuals Using Wheelchairs*. UT-21.10, Utah Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/56982>
- [3] Schultz, G. G., **Macfarlane, G. S.**, Wang, B. & McCuen, S. (2020). *Evaluating the Quality of Signal Operations Using Signal Performance Measures*. UT-20.08, Utah Department of Transportation. <https://rosap.ntl.bts.gov/view/dot/54639>
- [2] Zalewski, A., Sonenklar, D., Cohen, A., Kressner, J. & **Macfarlane, G.** (2019). *Public Transit Rider Origin–Destination Survey Methods and Technologies*. TCRP Synthesis 138, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine. <https://doi.org/10.17226/25428>
- [1] Cruz, J., **Macfarlane, G.**, Xu, Y., Rodgers, M. O. & Guensler, R. (2015). *Sustainable Transportation Curricula*. National Center for Sustainable Transportation. <https://escholarship.org/uc/item/3c13q43c>

Conferences & invited talks

Conferences

P: paper/presentation, M: moderator, C: organizing committee

- 2026 Intermountain Engineering and Technology Conference, hEART European Association for Research in Transportation Annual Symposium, 17th International Conference on Ambient Systems Networks and Technologies (ANT)
- 2025 Elsevier Transportation Research Symposium, Modeling Mobility (MoMo)[™], International Conference on Transportation & Development (ICTD), For the Benefit of the World Conference
- 2024 Utah Transportation Conference, Activity-based Modeling Symposium[®], International Conference on Travel Behavior Research (IATBR), 15th International Conference on

- Ambient Systems Networks and Technologies (ANT), Transportation Research Board Annual Meeting (TRB)
- 2023 Innovations in Transportation Analysis and Planning^{en}, Move Utah Summit, Utah Transportation Conference
- 2022 Mobil.TUM, Transportation Research Board Annual Meeting (TRB)
- 2021 North American Regional Science Conference (NARSC)
- 2020 Transportation Research Board Annual Meeting (TRB)
- 2015 North American Regional Science Conference (NARSC)
- 2014 North American Regional Science Conference (NARSC)
- 2013 North American Regional Science Conference (NARSC)

Invited talks

- 2025 California HSR model transferability, Presentation, CalTrans Transit Research-to-Practice Symposium
- 2023 Modeling with Big Data, course presentation, Technische Universität München

Committee work & voluntary roles (selection)

- Memberships IPSA, ISA, APSA, EPCR
- Reviewer Zeitschrift für Politikwissenschaft, European Journal of Political Research, Journal of Comparative Politics, Revista Brasileira de Ciência Política
- 01/2022-12/2023 Member of the Curriculum Development Committee, Department of Political Science, University of Excellence
- 09/2020-08/2024 Member of the Ethics Review Board, Institute for Political Science, University of Excellence

Teaching experience (Justus-Liebig-University Giessen)

Supervised theses

Bachelor 7 Master 5 Teacher program 8

Teaching (selected)

Term	Course	Study program
Summer 2024	Statistics+: Starting with R (de)	Bachelor
Winter 2023	Analyzing political support (de)	Master
Summer 2023	Empirical method essentials (de)	Bachelor
Course language: (de) German, (en) English		

Additional training (selection)

Methods training

- 2024 Bayesian modelling in the Social Sciences, An expensive Spring Seminar

- 2023 **Applied multilevel regression modelling**, The coolest PhD Summer School
- Didactics**
- 2019 **Conflict competence I + II**, University Alliance
- 2015-2017 **Certificate of Competence in Higher Education**, University Alliance

Further skills

	Skill	Level	Comment
Computer languages	R		longstanding experience, taught several courses
	Quarto / Markdown		used in presentations, dashboards & reports
	LaTeX / Typst		used in presentations and papers, first steps with own templates/packages
Software	Positron		first steps
	Office		longstanding experience (Win/macOS)
Languages	German		native tongue
	English		fluent in speech and writing
	Portuguese		advanced
		basic knowledge	extensive knowledge
		intermediate knowledge	expert knowledge